

Marathi Machine Translation via Bodhan Fine-Tune

AI4Bharat — AI Research Engineer take-home
English → Marathi (eng_Latn → mar_Deva)

Source code: <https://github.com/Kaustubh-Rathi/marathi-mt-bodhan>

Artifacts, checkpoints & logs: https://drive.google.com/drive/folders/1Abt7fldMDI_4pjTa3ayDbLI1dqDZSUTD

Primary model: bodhan-ai/indic-translate (Gemma-4 E4B, 8B), QLoRA

Train: ai4bharat/samanantar (config `mr`), 8,000 pairs

Eval (held out, never trained on): ai4bharat/IN22-Gen (test), facebook/flores eng_Latn-mar_Deva (devtest)

1. What was delivered

Three fine-tuning sessions, each run end-to-end on a Kaggle GPU, with a full resumable checkpoint saved and mirrored to Drive every 5 optimizer steps.

Session	Method	Steps	Status	GPU
A — primary: Bodhan 8B QLoRA	qlora	16		–
B — fallback: IndicTrans2 200M LoRA	lora-seq2seq	16		–
C — ablation (r 16/ α 32, LR 5e-5)	qlora	16		–

Session A is the assignment-aligned primary. Session B is an independent 200M fallback that guarantees a complete story regardless of what happens on the 8B. Session C is a reduced-adapter / lower-LR ablation of A. All three completed and were evaluated on the held-out sets below.

Evaluation — all three sessions, held-out sets

Session	IN22 BLEU	IN22 chrF2	FLORES BLEU	FLORES chrF2
A — Bodhan 8B QLoRA	7.81	24.04	9.64	28.61
B — IndicTrans2 200M LoRA	20.91	54.10	19.49	54.90
C — ablation (r 16/ α 32, LR 5e-5)	0.27	0.71	1.16	2.38

These runs are 20-step smoke runs that prove the pipeline end to end, not converged fine-tunes, and the table reads that way: the 200M model has moved furthest in 20 optimizer steps, the 8B has barely begun to shift, and halving the adapter while also lowering the learning rate leaves the ablation almost unmoved. I report them as-is rather than tuning for a prettier number — the useful conclusion is that step count, not configuration, is the binding constraint here, and that these numbers should not be read as a quality comparison between the models.

2. Key decisions and why

Decision	Choice	Reasoning
Base model	bodhan-ai/indic-translate (8B) as primary; IndicTrans2 200M as fallback	The 8B is the assignment-aligned pick and the stronger Marathi prior, but it is gated and must fit a T4. The 200M trains anywhere and de-risks the schedule, so it runs in parallel rather than as a fallback branch.
Fine-tune stack	TRL + PEFT + BitsAndBytes NF4	Transparent and auditable. Unsloth is faster but its patched kernels hide exactly the failures that needed diagnosing; Axolotl adds a layer between me and those errors.
LoRA target modules	<code>all-linear</code> minus vision/audio and <code>lm_head</code>	Gemma-4 uses <code>ClippableLinear</code> ; a bare <code>q_proj</code> list crashes adapter injection. This form is encoded in <code>base.yaml</code> .
Experiment tracking	TensorBoard + <code>experiments.csv</code> + Drive sync	Zero infrastructure. A hosted tracker was rejected because it adds account and egress friction across three Kaggle accounts for three runs.
Compute	3 Kaggle accounts, one session each	A single account's 30h/week means one failure ends the story. Each session streams checkpoints to Drive, so a 12h kill costs at most one checkpoint.
Optimizer	<code>paged_adamw_8bit</code> , no DeepSpeed	Paged optimizer absorbs T4 memory spikes. ZeRO-3 shards the frozen base out from under PEFT and is wrong for single-GPU LoRA.

3. Data, evaluation and decontamination

Dataset pivot. The preferred education-domain set was access-gated and I was denied, so training uses the open `ai4bharat/samanantar` (`mr`) slice, capped at 8,000 train / 1,000 dev pairs, seed 42.

Evaluation is on general-domain sets disjoint from the training slice — IN22-Gen (test) and FLORES devtest — so any gain has to come from transfer rather than leakage.

Decontamination (full L1–L10 table in `reports/LEAKAGE_CHECKLIST.md`): pair-hash blocklist of the test sets against train, exact and source-only near-duplicate filtering at 0.9 similarity, 1–100 word length filters, and provenance checks. Scoring decoding is fixed at 256 new tokens, beam 5.

4. Engineering log: four failures, diagnosed not worked around

Each of these killed a run at the first optimizer step. All four are fixed in code, documented inline at the fix, and re-verified by a completed run.

1. **QLoRA OOM (10.5 GiB on a 16 GiB T4)**. PEFT's stock `prepare_model_for_kbit_training` upcasts every low-precision tensor to FP32, including Gemma's multi-gigabyte embedding table. Replaced with an architecture-aware helper that upcasts only 1-D norm vectors and keeps large matrices in the compute dtype.
2. **assistant_only_loss RuntimeError**. TRL aborts when a row has no assistant mask, because the gated chat template emits no generation markers. Disabled it for these runs and recorded the semantic consequence (full-sequence loss) rather than silently accepting it.
3. **GradScaler vs BF16**. On a T4, `torch.cuda.is_bf16_supported()` returns True through sm75 emulation, so the model loaded in BF16 while the trainer ran the FP16 scaler: `_amp_foreach_non_finite_check_and_unscale_cuda` not implemented for 'BFloat16'. Fixed by letting explicit config flags override the GPU capability probe instead of trusting the probe.
4. **FP16 master weights**. Forcing FP16 only moved the failure: `Attempting to unscale FP16 gradients`. The AMP contract needs FP32 master weights with autocast doing the casting, so trainable LoRA params are upcast to FP32 *after* the adapters are built — they are created inside `SFTTrainer.__init__`, so aligning earlier had missed all 100,999,168 of them. The frozen 4-bit base stays low-precision, keeping memory in budget.

5. Reproducing this

```
pip install -r requirements.txt
make data          # download + decontaminate + build train/dev/test
make train-a       # primary Bodhan 8B QLoRA
make train-b       # IndicTrans2 fallback
make eval          # IN22-Gen + FLORES -> reports/metrics.json
```

Training runs as Kaggle *script* kernels (no notebooks), one account per session, driven by `scripts/kaggle/push_kernel.ps1`. Every checkpoint is rcloned to Drive in the background as it is written, each verified with an `_upload_complete` marker before local copies are pruned; after a 12h kill, set `MR_MT_RESUME=auto` and re-push to resume from the last confirmed checkpoint.

6. Honest caveats

- Training data pivoted away from the preferred education set because access was denied, and Samanantar was part of IndicTrans2's pretraining corpus, so Session B's score partly re-measures memorisation.
- These are 20-step smoke runs, not converged fine-tunes. The metric table demonstrates the evaluation path works end to end; it is not a quality comparison between the models.
- Assistant-only loss is disabled, so the loss covers the full sequence rather than the assistant span alone.
- Session B logs gradient norms around $3.5e4$ against `max_grad_norm=1.0` — heavy clipping every step. A lower learning rate or longer warmup is the first thing to change for a real run.

Full detail lives in the repository: `APPROACH.md` (narrative and rejected alternatives), `docs/HYPERPARAMETERS.md` (every knob and its rationale), `docs/SPEC.md`, `docs/SETUP.md`, `docs/TRAINING.md`, and `reports/` for live artifacts.